

EMS — Event Management System

A full-stack web application that manages conference room bookings between **investors** and **presenters**, matching them by shared sector interests, time availability, and hotel room availability.

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What is EMS?

EMS is a conference booking platform designed for events where **presenters** (representing companies) meet with **investors** to pitch their business. Every meeting takes place inside a **conference room** at a **hotel**, lasts exactly **one hour**, and can only be booked when all three parties — investor, presenter, and room — are free at the same time and share the same sector of interest.

How a Reservation Works

The reservation flow follows a strict matching process to ensure every booking is valid:

1. Select Investor
↓
2. Choose Sector → Pick Date → Pick Available Time Slot
↓ (only dates/times where investor has availability + free rooms are shown)
3. Pick a Presenter
↓ (only presenters who share the sector and are free at that time appear)
4. Pick a Room & Confirm
↓ (only unbooked rooms at that hotel/date/time are shown)
5. Booking Confirmed ✓

Once confirmed, the system **marks the investor, presenter, and room as occupied** for that time so no double-booking can occur.

Pages & Features

Home

The landing page with quick links to:

- Start a new reservation
 - Access the admin panel (Hotels, Investors, Presenters)
-

Hotels

Manage all hotel and conference room information.

What you can do:

- Add a new hotel (name + address)
- Add conference rooms to each hotel
- Add time slots to each conference room (e.g., 1PM–4PM creates individual 1-hour slots: 1–2PM, 2–3PM, 3–4PM automatically)
- Delete hotels, rooms, or individual time slots — deletions are permanent and cascade properly

Key behaviours:

- Time slots are stored as individual 1-hour blocks in the database
 - Deleting a room removes all its slots and any reservations tied to those slots
 - Deleting a hotel removes all its rooms, slots, and related reservations
-

Investors

Manage investor profiles and their sector availability windows.

What you can do:

- Add investors with name and mobile number
- Define which sectors an investor is interested in, and the date + start time for each session
 - End time is automatically set to 1 hour after the selected start time
- Filter and search investors by name, mobile, or sector
- Delete investors (also removes their reservations automatically)

Example:

Investor "CIB" is available in the **Finance** sector on June 21 starting at **1PM** (system sets end to 2PM).

Presenters

Manage presenter profiles in exactly the same way as investors.

What you can do:

- Add presenters with name and mobile
- Define sector availability with date and start time (end time is auto +1 hour)
- Filter and search by name, mobile, or sector
- Delete presenters (also removes their reservations automatically)

Example:

Presenter "Oltob" is available in **Restaurants** on June 21 starting at **1PM**.

Reservation Wizard

The core feature of EMS — a 4-step booking flow that intelligently matches investor + presenter + room.

Step 1 — Select Investor

Browse the full investor list. Use the search bar to filter by name or sector. Click to select.

Step 2 — Choose Sector, Date & Time

- **Sector toggles** appear based on what the selected investor is interested in
- The **date picker** highlights available days in green (investor has availability + free rooms exist) and unavailable days in red
- **Time slots** appear after picking a date — only hours that overlap with the investor's availability window AND have at least one free room are shown

Step 3 — Pick a Presenter

Only presenters who:

- Share the same sector
- Are available on the same date and time
- Have not already been booked at that slot

...appear in this list.

Step 4 — Confirm Room

Only hotel rooms that:

- Have a free time slot on the selected date and hour
- Have not been booked by anyone else

...are shown. Select a room and confirm to lock the booking.

After confirmation: A success screen shows the full booking summary (investor, presenter, sector, hotel, room, date, and time).

Report

A read-only view of all confirmed reservations, showing:

- Investor name
- Presenter name
- Sector
- Hotel and conference room
- Date and time of the meeting
- Booking timestamp

The report is also available as an **SSRS report** connected directly to the SQL Server database for printable/exportable output.

Business Rules

Rule	Details
Meeting duration	Always exactly 1 hour
Matching condition	Investor and presenter must share the same sector and be free at the same date and time
No double-booking	An investor, presenter, or room can only have one booking per time slot
Cascade delete	Deleting an investor, presenter, hotel, room, or slot removes all related reservations
Time slot creation	Entering a range (e.g., 1PM–4PM) auto-creates individual 1-hour slots
Availability entry	Only a start time is needed; end time is auto-calculated as +1 hour

Data Import — SSIS Package

Hotels can be imported in bulk using the included **SSIS (SQL Server Integration Services) package**.

How it works:

- The package reads multiple CSV files from a configured folder at once
- Each CSV file contains one hotel record per line
- All hotels are inserted into the database automatically
- Useful for seeding large numbers of hotels without using the UI

CSV format:

```
HotelName,Address
Hilton Cairo,123 Nile Street
Four Seasons,456 Downtown Blvd
```

Tech Stack

Layer	Technology
Frontend	Angular 17+ (Standalone Components, Signals)
Styling	SCSS with Angular Material
Backend	ASP.NET Core (.NET 10) — REST API
Database	SQL Server (EMS_DB)
ORM	Entity Framework Core
Matching Logic	SQL Stored Procedures
Reporting	SSRS (SQL Server Reporting Services)
Data Import	SSIS (SQL Server Integration Services)

Getting Started

Prerequisites

- [.NET 10 SDK](#)
- [Node.js 20+](#)
- SQL Server (local or remote)
- Angular CLI (`npm install -g @angular/cli`)

1 — Database Setup

Run the SQL scripts in order from [DataOps/Database/](#):

```
-- 1. Create tables
01_CreateTables.sql

-- 2. Create stored procedures
02_StoredProcedures.sql

-- 3. Seed base data (6 sectors)
03_SeedData.sql
```

2 — Backend

```
cd Backend/EMS.API
dotnet run
# API runs at: http://localhost:5000
```

Update the connection string in `appsettings.Development.json` to point to your SQL Server instance.

3 — Frontend

```
cd frontend/redesign
npm install
npm start
# App runs at: http://localhost:4200
```

Demo

 Videos demonstrating the full reservation flow and admin panel are included with this submission.